

Practice Problem Sheet-3
Transform Calculus
(MA20202)

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Q1. Solve $(D^2+1)y = t \cos 2t$ $y(0)=0, y'(0)=0$

Ans: $y = -\frac{5}{9} \sin t + \frac{4}{9} \sin 2t - \frac{t}{3} \cos 2t$

Q2. Solve $y''(t) + 2y'(t) + y(t) = 2e^{1-t}$, $y(1)=1, y'(1)=-1$

[Hint: Find $Y(s)$ with c_1 & c_2 , then use initial condition
 $y(t) = e^{1-t}(t^2 - 2t + 2)$]

Q3. Solve $y'' - 6y' + 9y = t^2 e^{3t}$ $y(0)=2, y'(0)=17$

Ans: $y(t) = 2e^{3t} + 11te^{3t} + \frac{1}{12}t^4 e^{3t}$

Q4. Solve $y'' + 4y' + 6y = 1 + te^{-t}$ $y(0) = y'(0) = 0$

Ans: $y(t) = \frac{1}{6} + \frac{1}{3}e^{-t} - \frac{1}{2}e^{-2t} \cos \sqrt{2}t - \frac{\sqrt{2}}{3}e^{-2t} \sin \sqrt{2}t$

Q5. Solve $(D+2)^2 y = 4e^{-2t}$ $y(0)=1, y'(0)=4$

Ans: $y(t) = e^{-2t}(2t^2 + 2t - 1)$

Q6. Solve $\frac{d^2 y}{dt^2} + 9y = \cos 2t$ $y(0)=1, y(\frac{\pi}{2})=-1$

Ans: $y(t) = \frac{1}{5} \cos 2t + \frac{4}{5} \cos 3t + \frac{4}{5} \sin 3t$

Q7. Solve $y'' + ty' - 2y = 4$ $y(0)=-1, y'(0)=0$

Ans: $y(t) = t^2 - 1$; Hint: After $Y(s)$, apply $Y(s) \rightarrow 0$ as $s \rightarrow \infty$.

Q8. Solve $y''(t) + y'(t) = t$ $y(1)=1, y'(1)=0$

Ans: $y(t) = \frac{(t-1)^2}{2} + 1$

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